

ECE 253 – Digital Image Processing

Image Denoising

Image Denoising

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Content Type

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Year

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1930

2016

From

To

1930

2016

☐ Comprehensive Analysis on the Effects of Noise Estimation Strategies on Image Noise Artifact Suppression Performance

Leigh, A.; Wong, A.; Clausi, D.A.; Fieguth, P.

Multimedia (ISM), 2011 IEEE International Symposium on

Year: 2011

Pages: 97 - 104, DOI: 10.1109/ISM.2011.24

Cited by: Papers (1)

IEEE Conference Publications

Abstract (html) PDF (640 Kb) CC

☐ Evolving Image Noise Filters through Genetic Programming

Banks, E.R.; Agarwal, P.; McBride, M.; Owens, C.

DoD High Performance Computing Modernization Program Users Group

Conference (HPCMP-UGC), 2009

Year: 2009

Pages: 307 - 312, DOI: 10.1109/HPCMP-UGC.2009.50

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IEEE TRANSACTIONS ON IMAGE PROCESSING, VOL. 19, NO. 4, APRIL 2010

895

Is Denoising Dead?

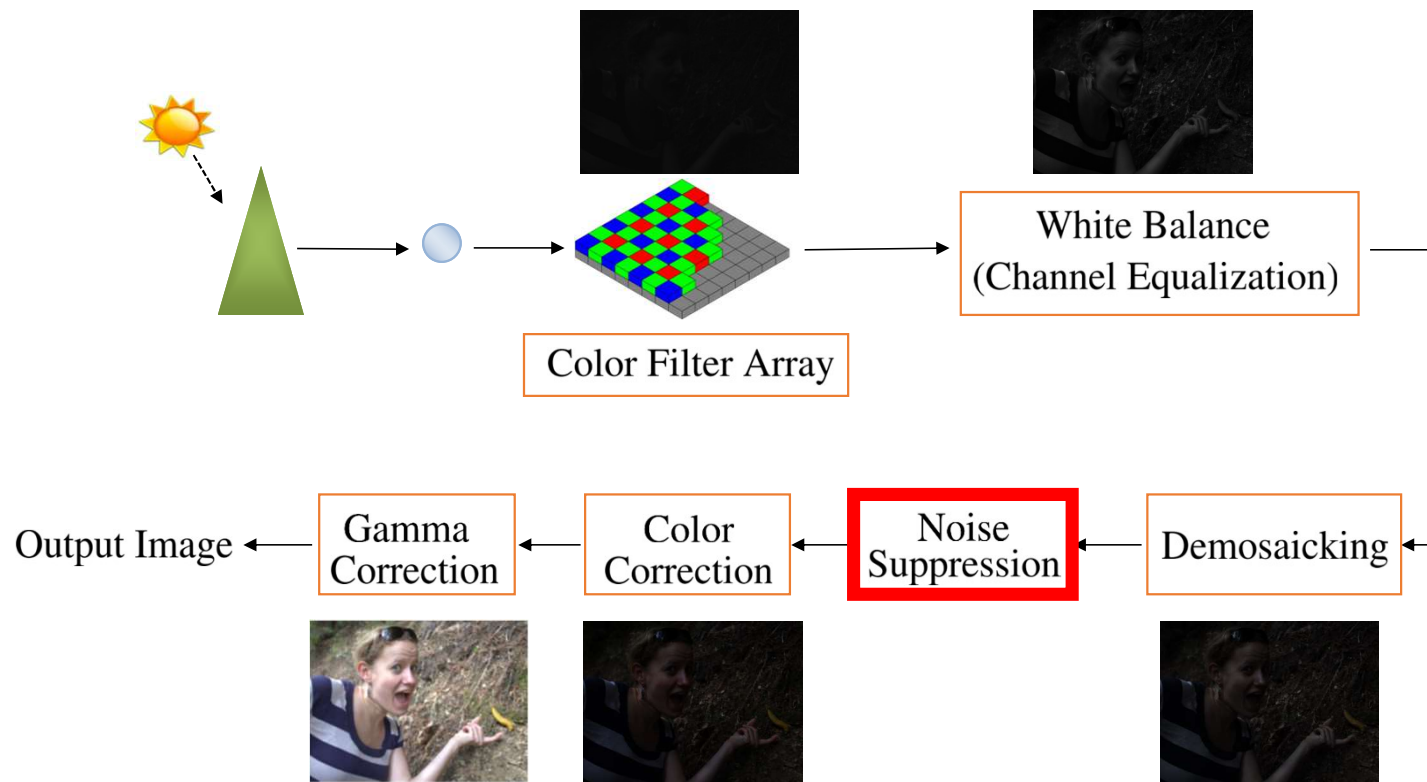
Priyam Chatterjee, *Student Member, IEEE*, and Peyman Milanfar, *Fellow, IEEE*

Abstract—Image denoising has been a well studied problem in the field of image processing. Yet researchers continue to focus attention on it to better the current state-of-the-art. Recently proposed methods take different approaches to the problem and yet

erature on such performance limits exists for some of the more complex image processing problems such as image registration [7], [8] and super-resolution [9]–[12]. Performance limits to object or feature recovery in images in the presence of point-

A Short Tour to Image Noise

Typical Camera Pipeline



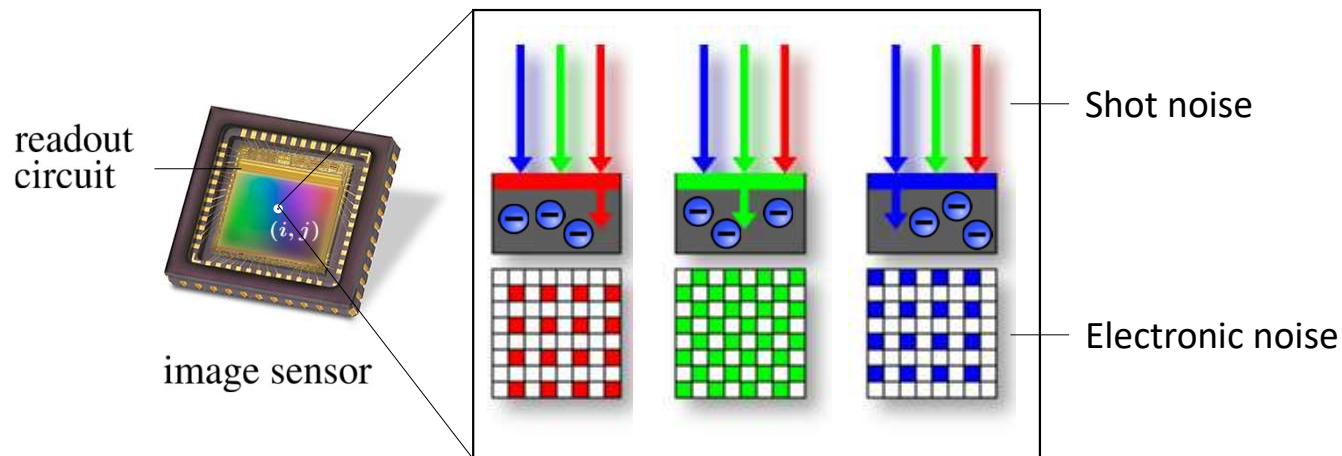
Sources of Noise

(1) Shot Noise

- Result of random photon arrival
- Poisson distributed
- Serious in low-light condition
- Not so bad under good light

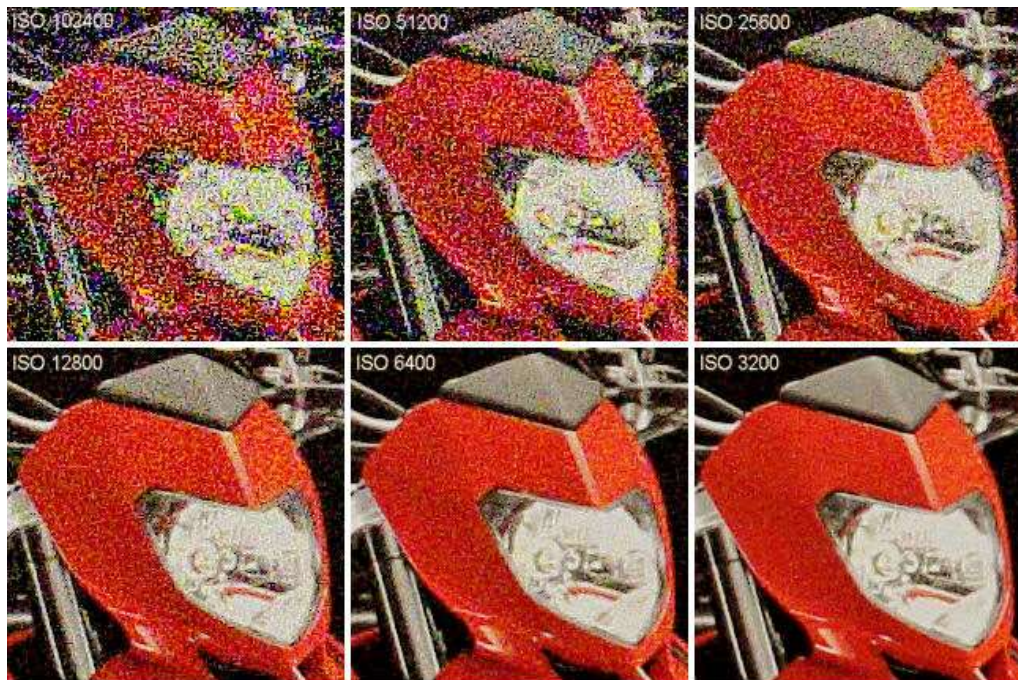
(2) Electronic Noise

- Instability of voltage/current
- Temperature fluctuation
- Analog to digital error
- Gaussian distributed



Simplified diagram illustrating the two sources of noise

Noise!



[<http://photo.net/equipment/canon/eos6D/review>]

Gaussian Noise!

Image Restoration



Physical limitations
Environmental factors



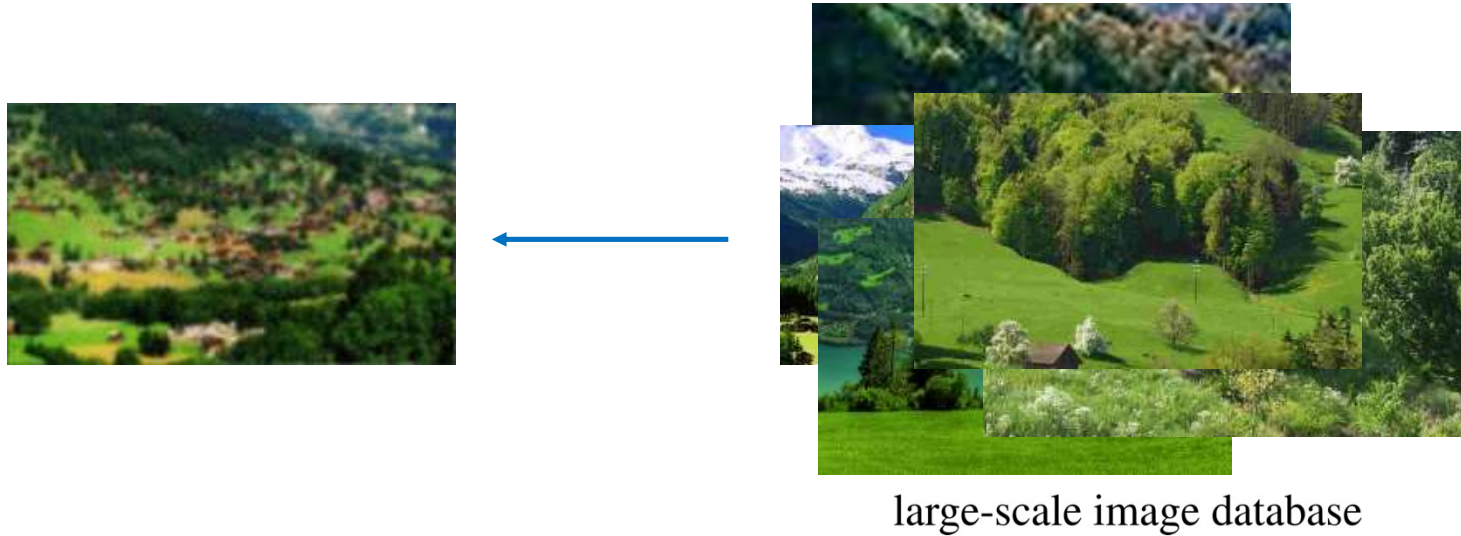
a degraded image
(with blur and noise, etc)

image restoration



a restored image
(sharp and clean)

Image Restoration: Where is it heading to?



New Opportunity:

Abundant images online are utilized for image restoration
(*e.g.*, learn a patch prior distribution from thousands of
patches sampled from a large database)

Large-scale Image Restoration

Challenges:

- Computational complexity
 - Time-consuming: large-scale of the database and inefficiency of the existing algorithms
- Estimation quality
 - Less effective: [Mairal et al., CVPR'09][Zoran et al., ICCV'11]

————→ New restoration methods are needed!

Image Denoising (A subclass of Image Restoration)



a noisy image
(no blur)

Image denoising:

restore a clean image from its noisy observation

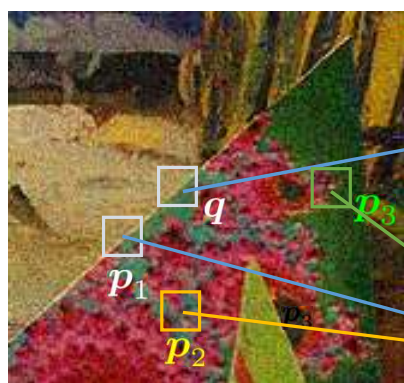
Outline

- Patch-based Image Denoising
- Database Adaptivity
- Algorithmic Adaptivity
 - Patch Matching
 - Denoising Filter Design
- Experimental Results

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Patch-Based Image Denoising



$\mathbf{q} = \mathbf{p} + \boldsymbol{\eta} \in \mathbb{R}^d$: patch to be denoised
 $\boldsymbol{\eta} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$

$\mathbf{p}_1 \dots \mathbf{p}_k \in \mathbb{R}^d$: patches with different similarities

Denoising process:

$$\hat{\mathbf{p}} = \Phi(\mathbf{q}; \mathbf{p}_1, \dots, \mathbf{p}_k)$$

$\hat{\mathbf{p}}$ is the estimate of $\mathbf{p}$

$\Phi(\cdot)$ is some linear or non-linear function.

Choice of $\Phi(\cdot)$

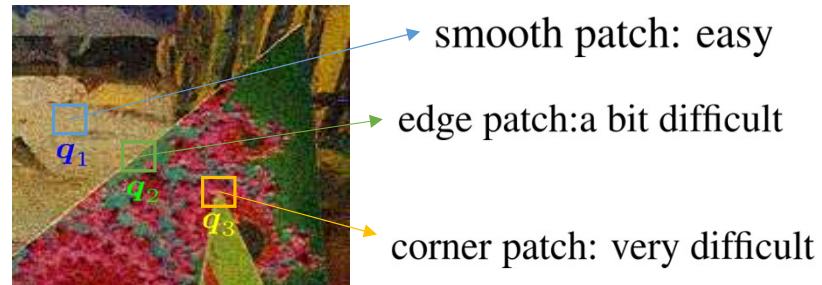
Non-local Means [Buades et al., SIAM'05]:

$$\hat{\mathbf{p}} = \frac{\sum_{i=1}^k \overbrace{e^{-\|\mathbf{q}-\mathbf{p}_i\|^2/h^2}}^{\text{weight for patch } \mathbf{p}_i} \mathbf{p}_i}{\underbrace{\sum_{i=1}^k e^{-\|\mathbf{q}-\mathbf{p}_i\|^2/h^2}}_{\text{for normalization}}}$$

$\Phi(\cdot)$ is a weighted averaging function

Sources of $p_1, \dots, p_k$

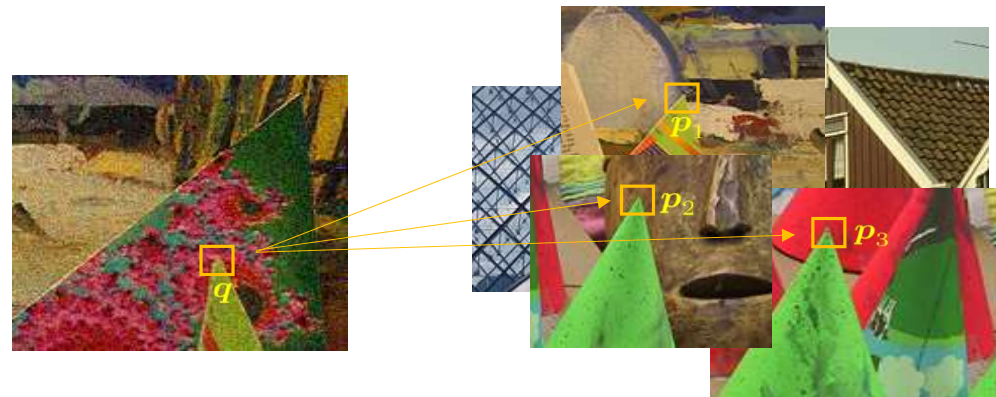
(a) noisy image itself:
internal



Drawback: limited performance

[Chatterjee and Milanfar, TIP'09, TIP'11]

(b) a database:
external



Drawback: time-consuming and marginal improvement

[Mairal et al., CVPR'09][Zoran et al., ICCV'11]

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Image Retrieval

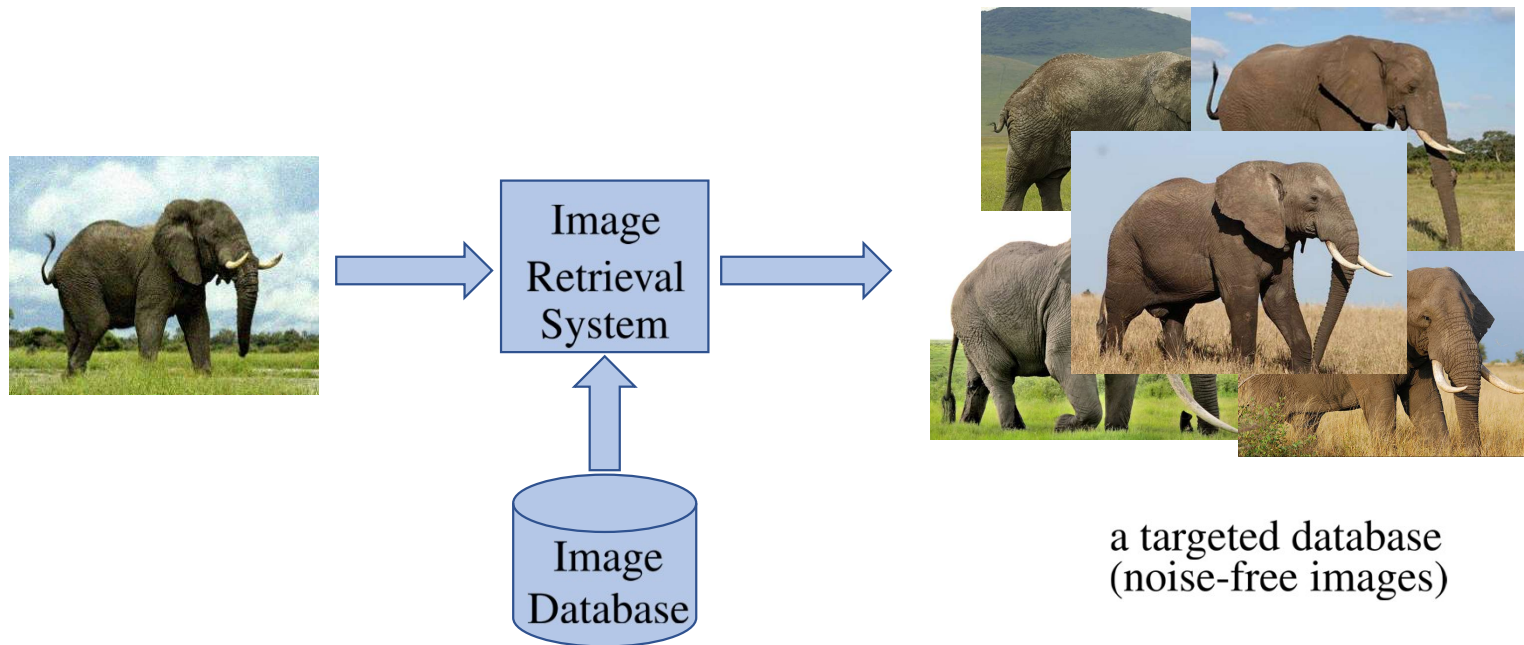
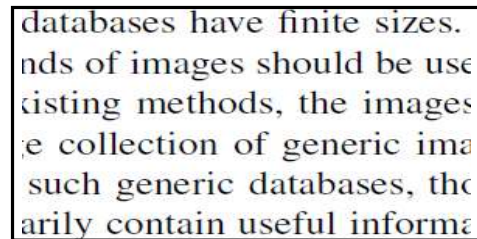


Image denoising using a targeted database!

Focus of this presentation: images that have reasonable amount of example images.



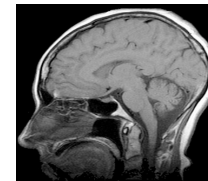
Text images(e.g., newspapers and documents)



Human faces



Images of license plates

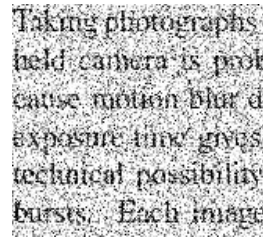


Medical CT and MRI images



Images of landmarks

Targeted Database: How to utilize it?



noisy image



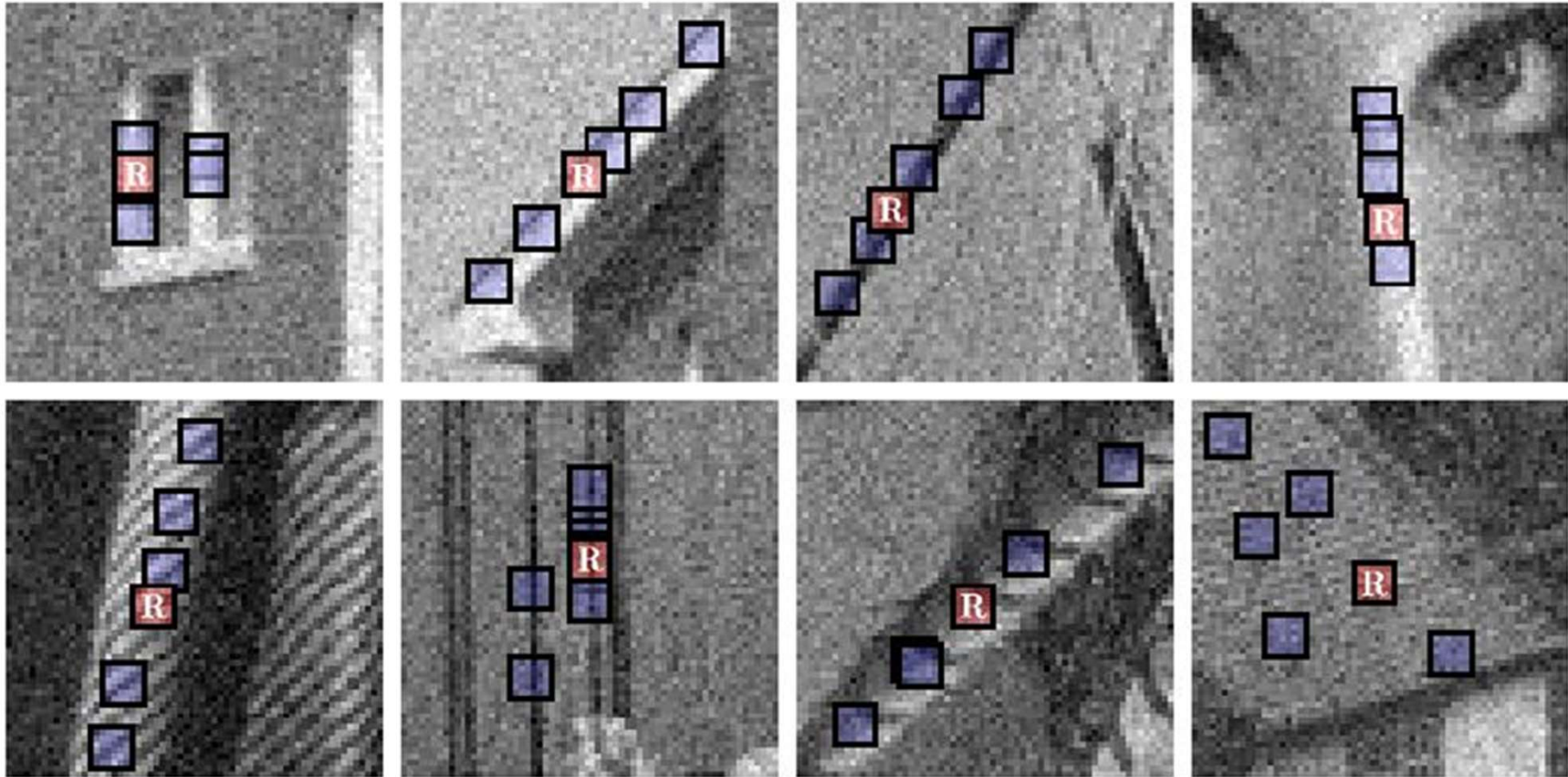
was motivated by modeling at a circuit level, it disagrees with [19]. In the latter work, however, the noise measurements are made not on the image sensors themselves, but rather on the output images from a digital camera that has already undergone the process of demosaicing. In this work, we assume that where x is the observed image, s is the ideal noiseless image, k_0 and k_1 are constants, and η represents white Gaussian noise with variance 1. Through experimentation, we verified that (1) is a good noise model for Agilent Technologies camera evaluation board HDCP-2000 equipped with 300 K pixel CMOS sensor and capturing raw sensor data. Left graph of Fig. 1 shows the noise standard deviation plotted against the sensor value. As predicted in (1), the graph clearly implies an affine line. While white Gaussian noise in (2), removes the signal-dependent noise component, it is not a good model for the noise in the generalized models. See Appendix I for a review of *general total variation* models and their applications. The noise model taken by translating image patches in the vicinity of s_0 . This approach is cited

targeted database
(noise-free images)

Two questions:

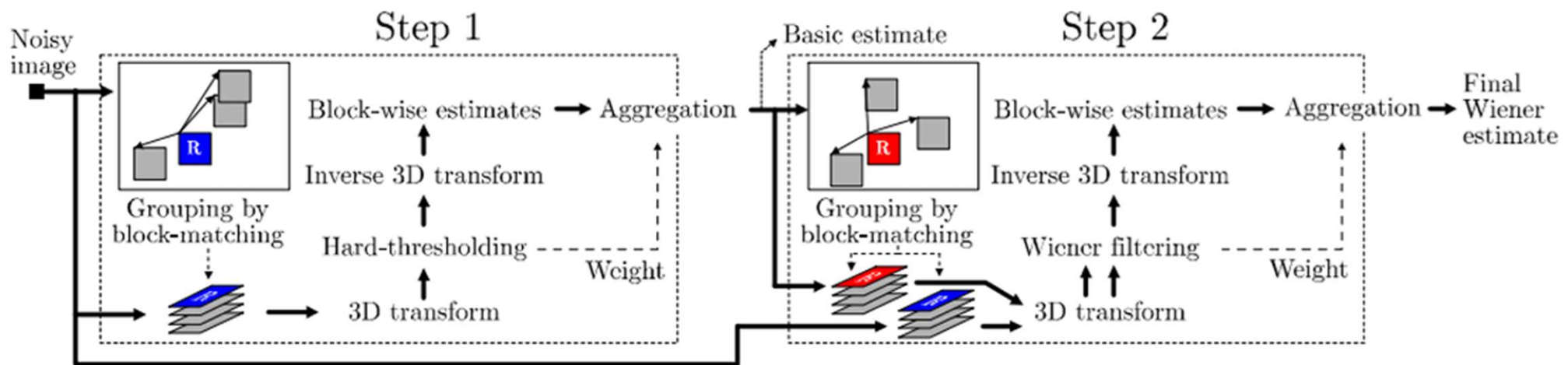
- How to find reliable reference patches?
- How to design effective and efficient algorithms?

BM3D



Dabov, Kostadin, et al. "Image denoising by sparse 3-D transform-domain collaborative filtering." IEEE Transactions on image processing 16.8 (2007): 2080-2095.

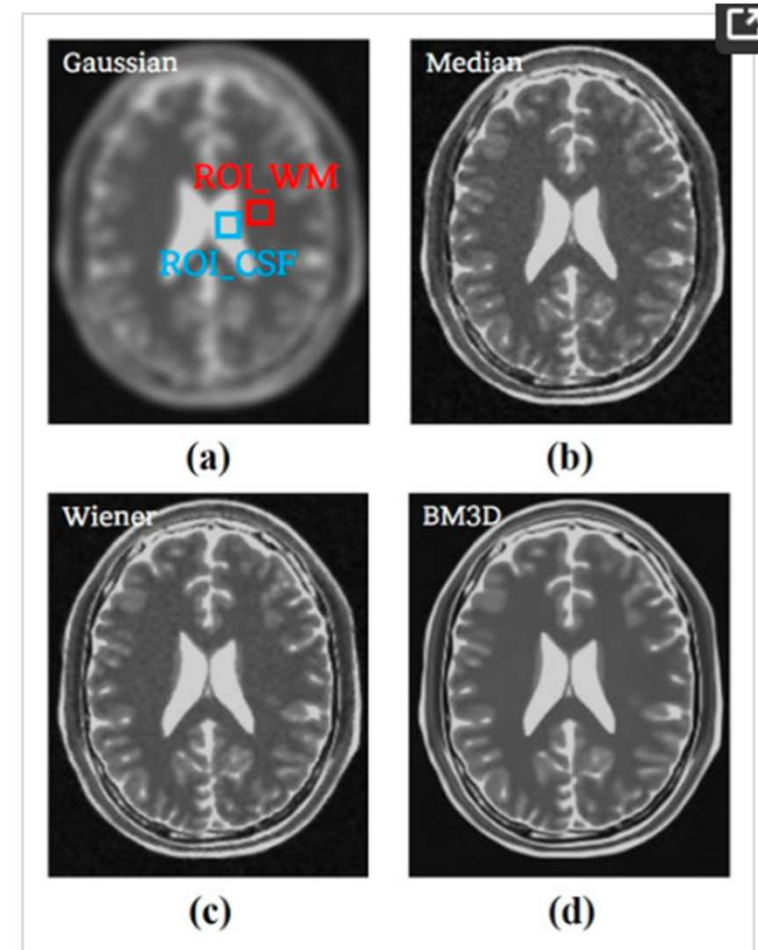
BM3D

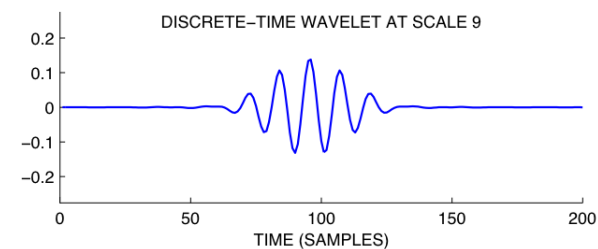
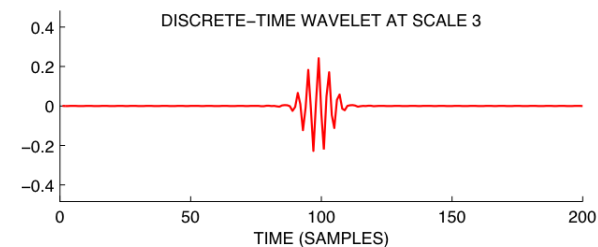
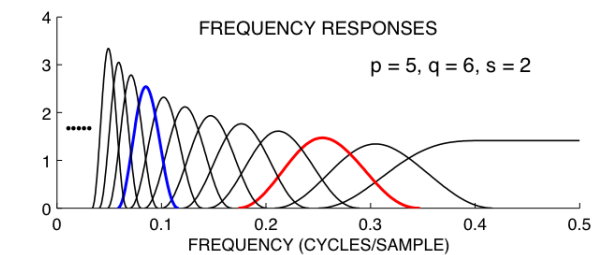
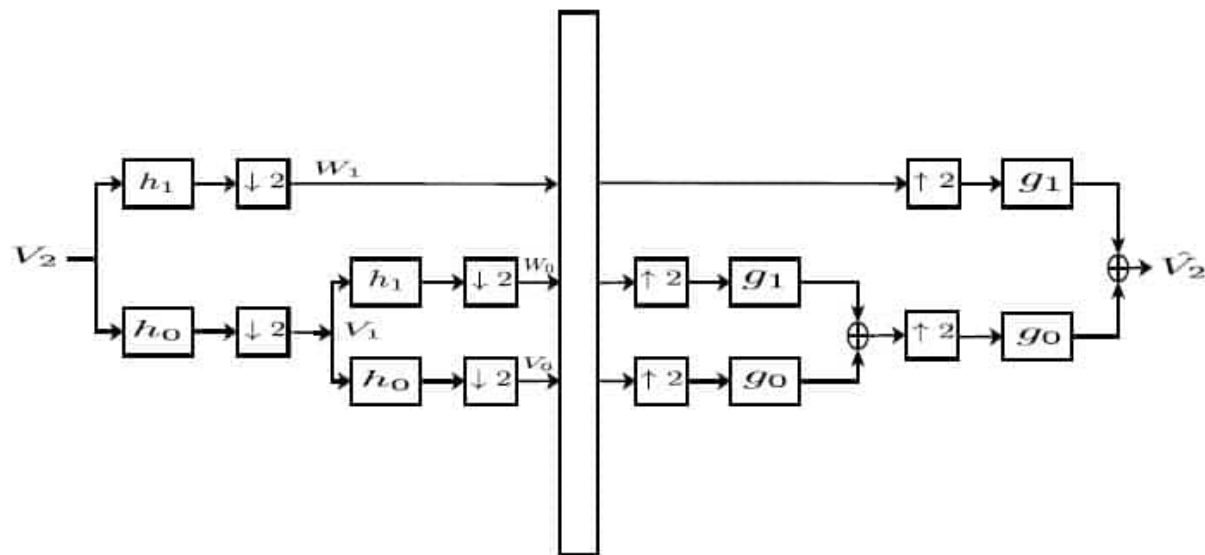


BM3D

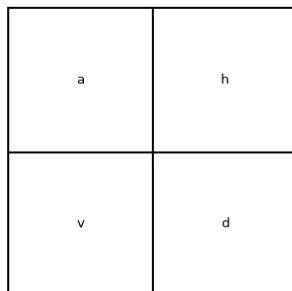


<https://biii.eu/bm3d-denoising>

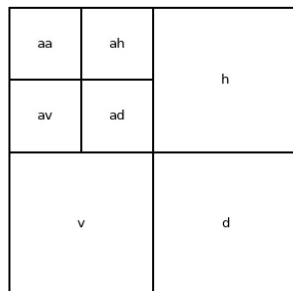




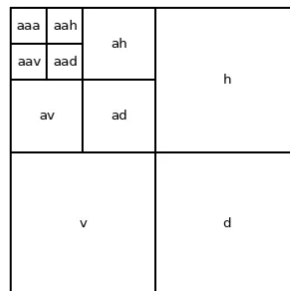
1 level decomposition



2 level decomposition



3 level decomposition

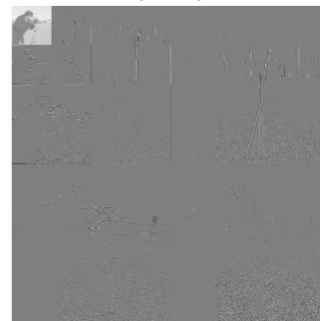
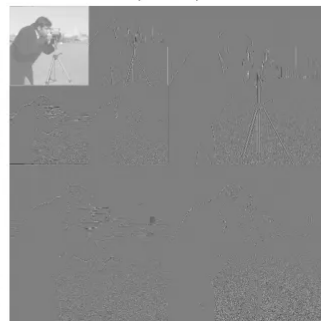


Image

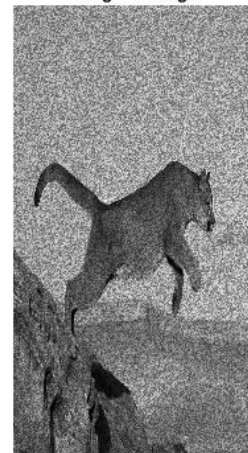
Coefficients (1 level)

Coefficients (2 level)

Coefficients (3 level)



Original Image



Denoised Image



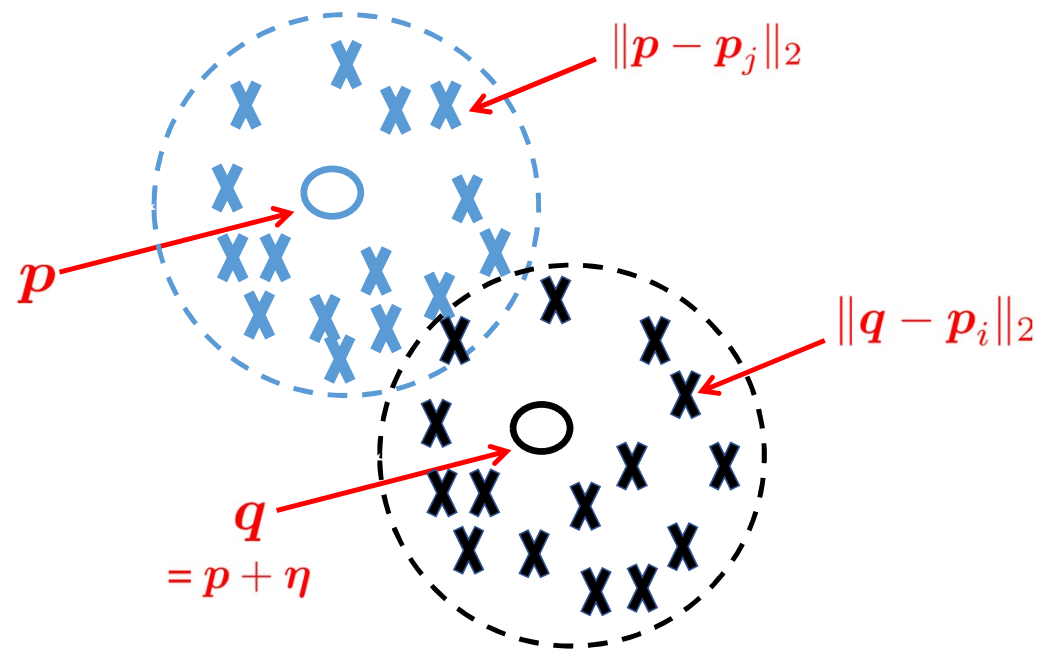
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Patch Matching: k nearest neighbors (kNN)



Patch Matching: k nearest neighbors (kNN)



Can we refine the patch matching?

kNN: Mathematical Formulation for Improvement

Problem formulation: search k nearest neighbors from n candidate patches

$$\begin{array}{ll}\underset{\mathbf{x}}{\text{minimize}} & \mathbf{c}^T \mathbf{x} \\ \text{subject to} & \mathbf{x}^T \mathbf{1} = k \\ & \mathbf{x} \in \{0, 1\}^n,\end{array}$$

where $\mathbf{c} \stackrel{\text{def}}{=} [c_1, \dots, c_n]^T$ with $c_i = \|\mathbf{q} - \mathbf{p}_i\|_2$.



$$\underset{\mathbf{x} \in \{0,1\}^n, \mathbf{x}^T \mathbf{1} = k}{\text{minimize}} \quad \sum_j x_j \|\mathbf{q} - \mathbf{p}_j\|_2$$

$$x_j = 1 \quad \implies \quad \|\mathbf{q} - \mathbf{p}_j\|_2 \text{ is small.}$$

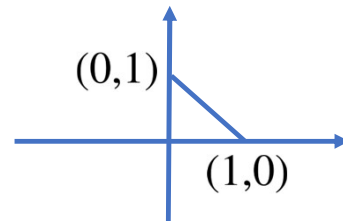
Solution is the same as k NN.

kNN: Relaxation

Problem formulation:

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & \mathbf{c}^T \mathbf{x} \\ \text{subject to} & \mathbf{x}^T \mathbf{1} = k \\ & \mathbf{x} \in \{0, 1\}^n. \end{array} \iff \begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & \mathbf{c}^T \mathbf{x} + \tau \varphi(\mathbf{x}) \\ \text{subject to} & \mathbf{x}^T \mathbf{1} = k \\ & 0 \leq \mathbf{x} \leq 1. \end{array}$$

e.g.: $n=2, k=1$



$$\mathbf{x} = [1, 0]^T, \text{ if } \|\mathbf{q} - \mathbf{p}_1\|_2 < \|\mathbf{q} - \mathbf{p}_2\|_2$$

$$\mathbf{x} = [0, 1]^T, \text{ if } \|\mathbf{q} - \mathbf{p}_2\|_2 < \|\mathbf{q} - \mathbf{p}_1\|_2$$

Remark: the new constraint has a convex set. Thus, as long as the objective function is convex, it could be solved by a convex toolbox.

Two Regularizations

Cross Similarity:

$$\begin{array}{ll}\underset{\mathbf{x}}{\text{minimize}} & \mathbf{c}^T \mathbf{x} + \tau \mathbf{x}^T \mathbf{B} \mathbf{x} \\ \text{subject to} & \mathbf{x}^T \mathbf{1} = k \\ & 0 \leq \mathbf{x} \leq 1\end{array}$$

Basic Estimate:

$$\begin{array}{ll}\underset{\mathbf{x}}{\text{minimize}} & \mathbf{c}^T \mathbf{x} + \tau \mathbf{e}^T \mathbf{x} \\ \text{subject to} & \mathbf{x}^T \mathbf{1} = k \\ & 0 \leq \mathbf{x} \leq 1\end{array}$$

Regularizations by Cross Similarity

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & \mathbf{c}^T \mathbf{x} + \tau \mathbf{x}^T \mathbf{B} \mathbf{x} \\ \text{subject to} & \mathbf{x}^T \mathbf{1} = k \\ & 0 \leq \mathbf{x} \leq 1, \end{array}$$

where $\mathbf{B} = [B_{ij}]$ with $B_{ij} \stackrel{\text{def}}{=} \|\mathbf{p}_i - \mathbf{p}_j\|_2$



$$\underset{0 \leq \mathbf{x} \leq 1, \mathbf{x}^T \mathbf{1} = k}{\text{minimize}} \quad \sum_j x_j \|\mathbf{q} - \mathbf{p}_j\|_2 + \tau \sum_{i,j} x_i x_j \|\mathbf{p}_i - \mathbf{p}_j\|_2.$$

$$x_i = 1 \text{ and } x_j = 1 \implies \begin{cases} \|\mathbf{q} - \mathbf{p}_i\|_2 \text{ and } \|\mathbf{q} - \mathbf{p}_j\|_2 \text{ are small.} \\ \|\mathbf{p}_i - \mathbf{p}_j\|_2 \text{ is small.} \end{cases}$$

This problem can be solved by convex toolbox.

Regularizations by Basic Estimate

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & \mathbf{c}^T \mathbf{x} + \tau \mathbf{e}^T \mathbf{x} \\ \text{subject to} & \mathbf{x}^T \mathbf{1} = k \\ & 0 \leq \mathbf{x} \leq 1, \end{array}$$

where $\mathbf{e} = [e_1, \dots, e_n]^T$ with $e_i \stackrel{\text{def}}{=} \|\bar{\mathbf{p}} - \mathbf{p}_i\|_2$

basic estimate



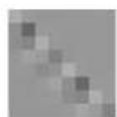
$$\underset{0 \leq \mathbf{x} \leq 1, \mathbf{x}^T \mathbf{1} = k}{\text{minimize}} \quad \sum_j x_j \|\mathbf{q} - \mathbf{p}_j\|_2 + \tau \sum_j x_j \|\bar{\mathbf{p}} - \mathbf{p}_j\|_2,$$

$$x_i = 1 \implies \begin{cases} \|\mathbf{q} - \mathbf{p}_i\|_2 \text{ is small.} \\ \|\bar{\mathbf{p}} - \mathbf{p}_i\|_2 \text{ is small.} \end{cases}$$

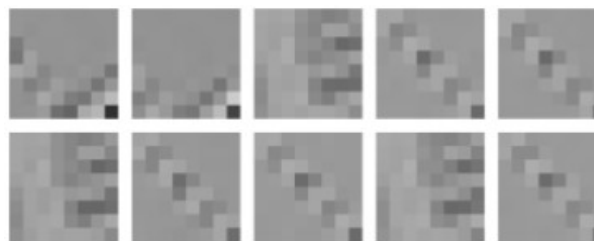
This problem can be easily solved.

Refined Patch Matching Results

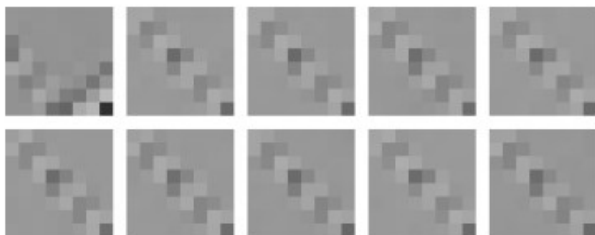
$k = 10, n = 200$



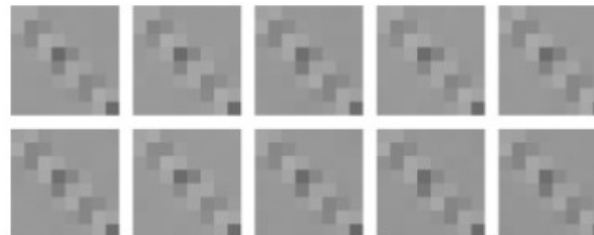
(a) p



(b) $\varphi(\mathbf{x}) = 0$
(k NN using noisy patch)



(c) $\varphi(\mathbf{x}) = \tau \mathbf{x}^T \mathbf{B} \mathbf{x}$
(regularized by cross similarity)



(d) $\varphi(\mathbf{x}) = \tau \mathbf{e}^T \mathbf{x}$
(regularized by basic estimate)

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Optimal Filter Design

Given:

$$\mathbf{q} = \mathbf{p} + \boldsymbol{\eta} \rightarrow \mathcal{N}(0, \sigma^2 \mathbf{I})$$

We want:

$$\mathbf{A} = \operatorname{argmin}_{\mathbf{A}} \mathbb{E} [\|\mathbf{A}\mathbf{q} - \mathbf{p}\|_2^2]$$

$$\longrightarrow (\mathbf{U}, \boldsymbol{\Lambda}) = \operatorname{argmin}_{\mathbf{U}, \boldsymbol{\Lambda}} \mathbb{E} \left[\left\| \underbrace{\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T}_{\text{eigen-decomposition on } \mathbf{A}} \mathbf{q} - \mathbf{p} \right\|_2^2 \right]$$

Optimal Filter Design

$$(\mathbf{U}, \mathbf{\Lambda}) = \operatorname{argmin}_{\mathbf{U}, \mathbf{\Lambda}} \mathbb{E} \left[\left\| \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T \mathbf{q} - \mathbf{p} \right\|_2^2 \right]$$

Lemma 1: *The optimal $\mathbf{U}$ and $\mathbf{\Lambda}$ are:*

$\mathbf{U} =$ any orthonormal matrix with the first column being $\mathbf{p}/\|\mathbf{p}\|$,

$$\mathbf{\Lambda} = \operatorname{diag} \left\{ \frac{\|\mathbf{p}\|^2}{\|\mathbf{p}\|^2 + \sigma^2}, 0, \dots, 0 \right\}.$$

Both $\mathbf{U}$ and $\mathbf{\Lambda}$ are not achievable, because they involve the ground truth $\mathbf{p}$.

Two questions:

- How to Find a Good U ?
- How to Find a Good Λ ?

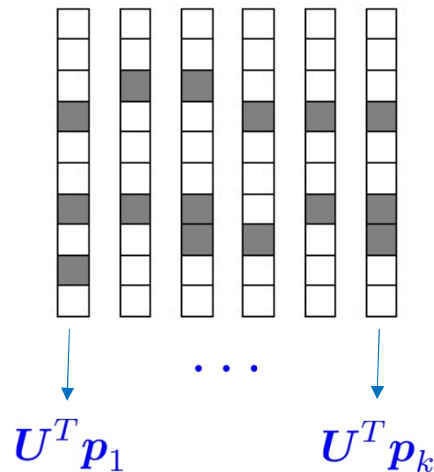
Two questions:

- How to Find a Good U ?
- How to Find a Good Λ ?

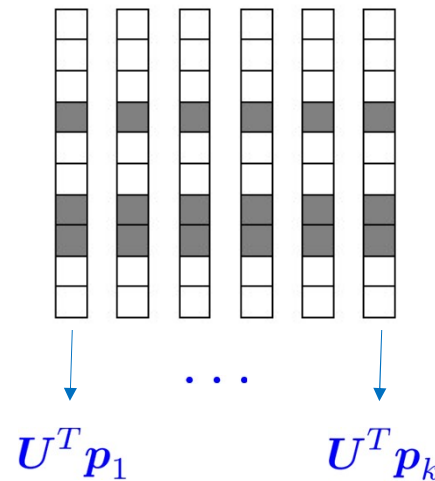
Learn U from the Reference Patches

Given the best k reference patches, $\mathbf{P} \stackrel{\text{def}}{=} [\mathbf{p}_1, \dots, \mathbf{p}_k]$, a good basis $\mathbf{U}$ should: (1) transform the patches to be similar in *magnitude* and *location*; (2) transform each patch to have as few non-zero entries as possible.

sparsity



group sparsity



Group Sparsity Minimization

Group sparsity minimization problem:

$$\begin{array}{ll} \underset{U}{\text{minimize}} & \|U^T P\|_{1,2} \\ \text{subject to} & U^T U = I. \end{array}$$

The $L_{1,2}$ norm computes the 2-norm of each row first and then 1-norm of the result.

Lemma 2 [Luo, Chan, and Nguyen 2014]: *The solution to the above minimization problem is:*

$$[U, S] = \text{eig}(PP^T),$$

where U is the eigenvector matrix, and S is the eigenvalue matrix.

Remark: PCA-based methods(e.g. [14],[15],[16]) are special cases of Lemma 2.

Weighted Group Sparsity Minimization

Unweighted group sparsity problem:

$$\begin{array}{ll} \underset{\mathbf{U}}{\text{minimize}} & \|\mathbf{U}^T \mathbf{P}\|_{1,2} \\ \text{subject to} & \mathbf{U}^T \mathbf{U} = \mathbf{I}. \end{array}$$

Solution:

$$[\mathbf{U}, \mathbf{S}] = \text{eig}(\mathbf{P}\mathbf{P}^T)$$

Weighted group sparsity problem:

$$\begin{array}{ll} \underset{\mathbf{U}}{\text{minimize}} & \|\mathbf{U}^T \mathbf{P} \mathbf{W}^{1/2}\|_{1,2} \\ \text{subject to} & \mathbf{U}^T \mathbf{U} = \mathbf{I}, \end{array}$$

$$\mathbf{W} = \frac{1}{\eta} \text{diag} \left\{ e^{-\|\mathbf{q} - \mathbf{p}_1\|^2/h^2}, \dots, e^{-\|\mathbf{q} - \mathbf{p}_k\|^2/h^2} \right\}$$

Solution:

$$[\mathbf{U}, \mathbf{S}] = \text{eig}(\mathbf{P} \mathbf{W} \mathbf{P}^T)$$

Two questions:

- How to Find a Good U ?
- How to Find a Good Λ ?

Optimal Λ when U is fixed

$$\Lambda = \underset{\Lambda}{\operatorname{argmin}} \mathbb{E} \left[\left\| U \Lambda U^T \mathbf{q} - \mathbf{p} \right\|_2^2 \right]$$

The optimal solution:

$$\Lambda = \operatorname{diag} \left\{ \frac{(\mathbf{u}_1^T \mathbf{p})^2}{(\mathbf{u}_1^T \mathbf{p})^2 + \sigma^2}, \dots, \frac{(\mathbf{u}_n^T \mathbf{p})^2}{(\mathbf{u}_n^T \mathbf{p})^2 + \sigma^2} \right\}.$$

It is not achievable, because it still involves the ground truth $\mathbf{p}$.

Bayesian Mean Squared Error (BMSE)

$$\mathbf{q} = \mathbf{p} + \boldsymbol{\eta}$$


$f(\mathbf{p})$ $\mathcal{N}(0, \sigma^2 \mathbf{I})$

$$\boldsymbol{\Lambda} = \underset{\boldsymbol{\Lambda}}{\operatorname{argmin}} \mathbb{E}_{\mathbf{p}} \left[\mathbb{E}_{\mathbf{q}|\mathbf{p}} \left[\left\| \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^T \mathbf{q} - \mathbf{p} \right\|_2^2 \mid \mathbf{p} \right] \right]$$

Lemma 3 [Luo, Chan, and Nguyen 2014]: *The optimal $\boldsymbol{\Lambda}$ to the above minimization problem is:*

$$\boldsymbol{\Lambda} = \frac{\operatorname{diag} \left\{ \mathbf{U}^T \boldsymbol{\mu} \boldsymbol{\mu}^T \mathbf{U} + \mathbf{U}^T \boldsymbol{\Sigma} \mathbf{U} \right\}}{\operatorname{diag} \left\{ \mathbf{U}^T \boldsymbol{\mu} \boldsymbol{\mu}^T \mathbf{U} + \mathbf{U}^T \boldsymbol{\Sigma} \mathbf{U} \right\} + \sigma^2 \mathbf{I}},$$

where $\boldsymbol{\mu} = \mathbb{E}(\mathbf{p})$ and $\boldsymbol{\Sigma} = \operatorname{cov}(\mathbf{p})$.

Learn μ and Σ from the Reference Patches

$$\Lambda = \frac{\text{diag} \left\{ U^T \mu \mu^T U + U^T \Sigma U \right\}}{\text{diag} \left\{ U^T \mu \mu^T U + U^T \Sigma U \right\} + \sigma^2 I}$$

$\mu \stackrel{\text{def}}{=} \underbrace{\sum_{i=1}^k \omega_i \mathbf{p}_i}_{\text{non-local mean}}$

$\Sigma \stackrel{\text{def}}{=} \underbrace{\sum_{i=1}^k \omega_i (\mathbf{p}_i - \mu)(\mathbf{p}_i - \mu)^T}_{\text{covariance of the patches}}$

We choose $\omega_i = \frac{1}{Z} e^{-\|\mathbf{q} - \mathbf{p}_i\|^2 / h^2}$ for some decay parameter h and normalization term Z

Efficient Computation of $\mathbf{U}$ and $\mathbf{\Lambda}$

Good News!

$\mathbf{U}$ and $\mathbf{\Lambda}$ can be *simultaneously* obtained through:

$$[\mathbf{U}, \mathbf{S}] = \text{eig}(\mathbf{P}\mathbf{W}\mathbf{P}^T),$$

and

$$\mathbf{\Lambda} = \frac{\mathbf{S}}{\mathbf{S} + \sigma^2 \mathbf{I}}.$$

Overall Efficient Algorithm

Algorithm 1 Proposed Algorithm

Input: noisy patch $\mathbf{q}$, and similar patches $\mathbf{p}_1, \dots, \mathbf{p}_k$

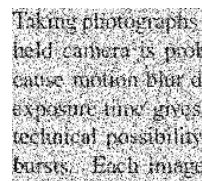
Output: estimate $\hat{\mathbf{p}}$

Learn $\mathbf{U}$ and $\mathbf{\Lambda}$

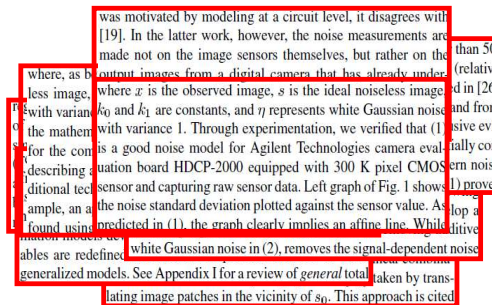
- Form data matrix $\mathbf{P}$ and weight matrix $\mathbf{W}$
- Compute eigen-decomposition $[\mathbf{U}, \mathbf{S}] = \text{eig}(\mathbf{P}\mathbf{W}\mathbf{P}^T)$
- Compute $\mathbf{\Lambda} = \frac{\mathbf{S}}{\mathbf{S} + \sigma^2 \mathbf{I}}$ (element-wise division)

Denoise: $\hat{\mathbf{p}} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T \mathbf{q}$

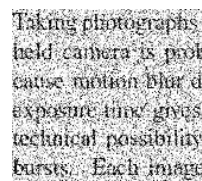
Constraint: Application Specific



noisy image



targeted database



noisy image



arbitrary database



Outline

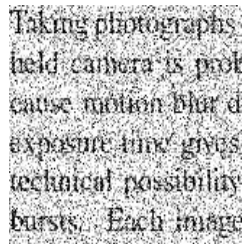
- Patch-based Image Denoising
- Database Adaptivity
- Algorithmic Adaptivity
 - Patch Matching
 - Denoising Filter Design
- Experimental Results

Experiments

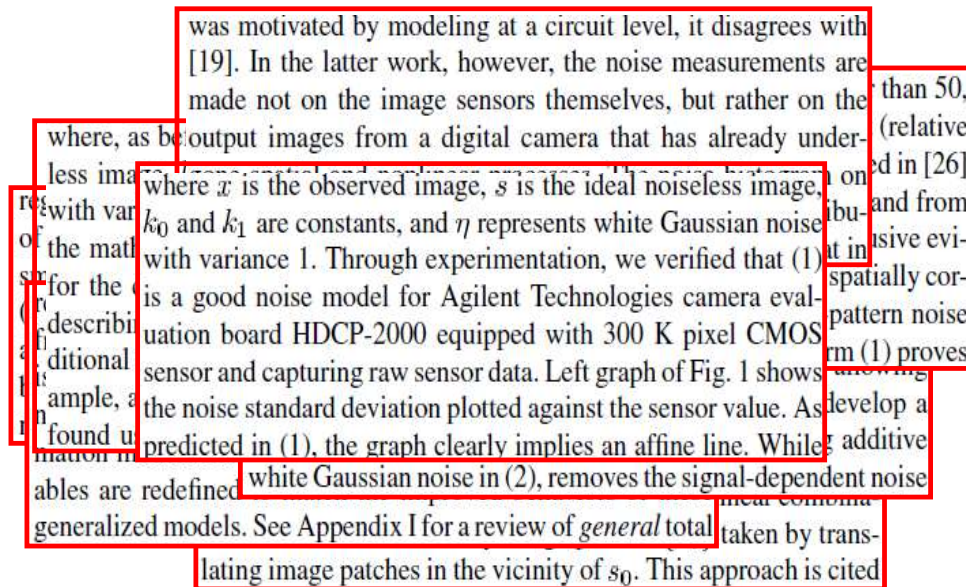
Methods to compare:

- Internal methods:
 - internal BM3D [Dabov et al., TIP'07]
- External methods:
 - external BM3D [Dabov et al., TIP'07]
 - external BM3D-PCA [Dabov et al., SPARS'09]
 - external LPG-PCA [Zhang et al., PR'10]
 - external NLM [Buades et al., SIAM'05]
 - external EPLL [Mairal et al., CVPR'09]

Experiment 1: Denoise text and documents

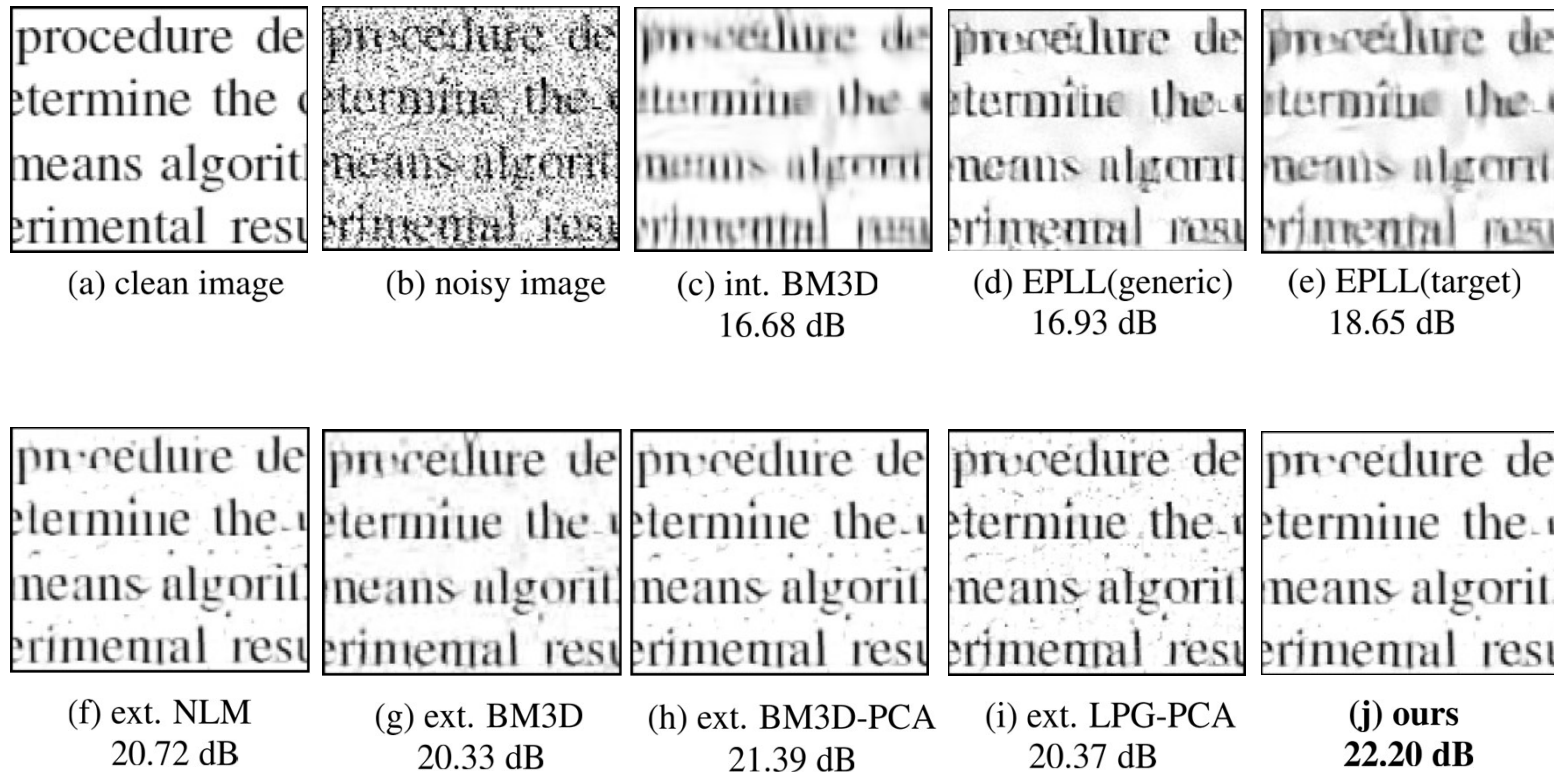


noisy text image



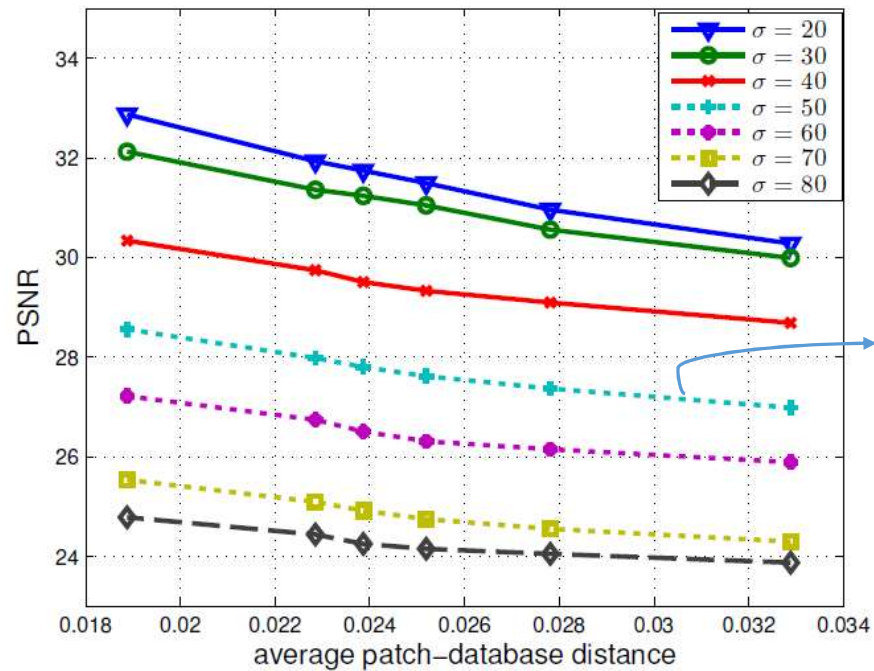
database with clean text images of the same scale

Experiment 1: Denoise text and documents



Visual and PSNR comparison

Experiment 1: Denoise text and documents



As database quality improves,
so does denoising performance.

Denoising performance in terms of the database quality

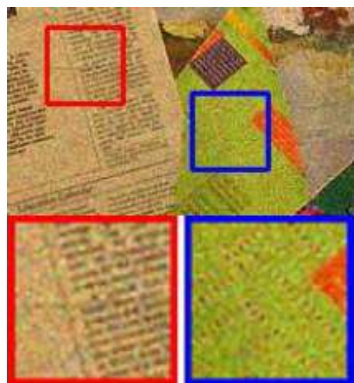
Experiment 2: Denoise Multi-view Images



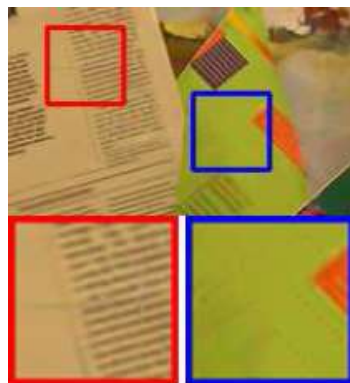
one noisy view due to the
dysfunction of one camera

a clean view from a different camera

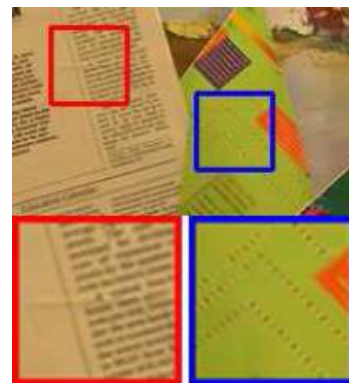
Experiment 2: Denoise Multi-view Images



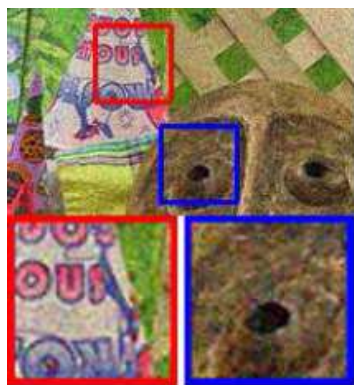
(a) noisy view (std:20)



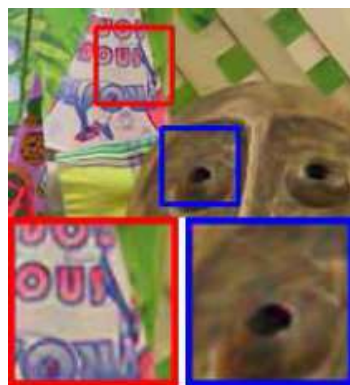
(b) iBM3D: 28.99 dB



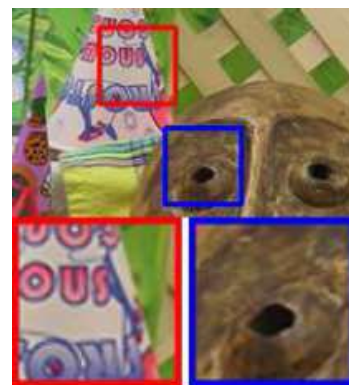
(c) **ours: 33.65 dB**



(d) noisy view (std:20)



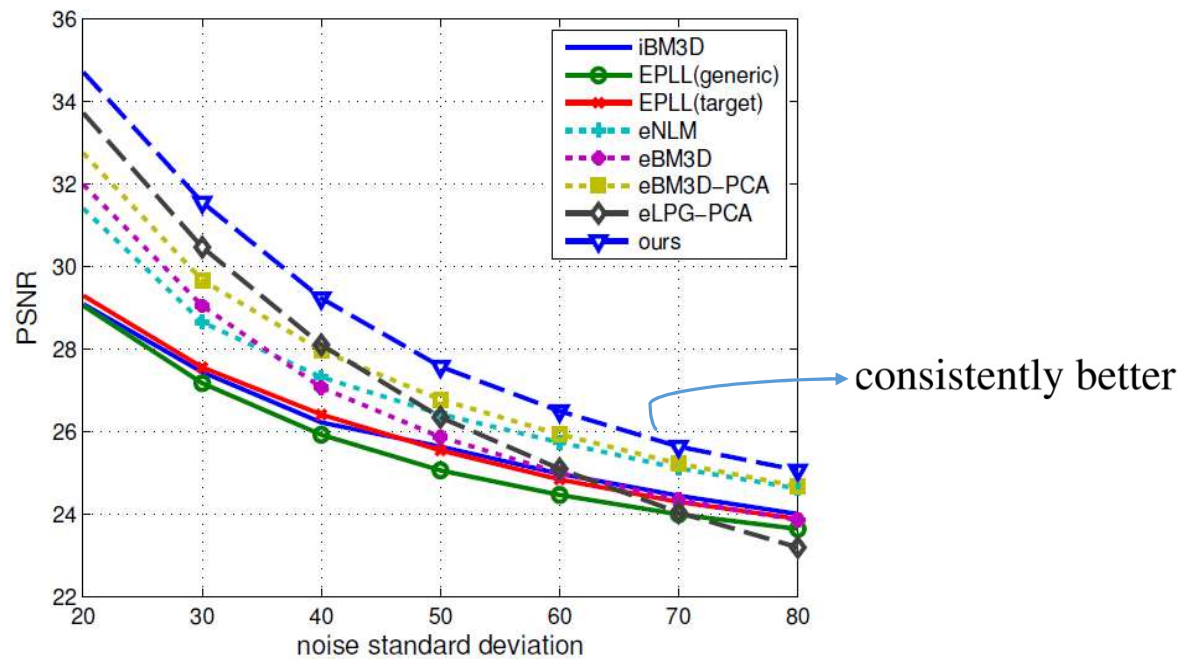
(e) iBM3D: 28.77 dB



(f) **ours: 32.11 dB**

Comparison with iBM3D (internal method)

Experiment 2: Denoise Multi-view Images



Multiview image denoising for “Barn”: PSNR vs noise levels

Experiment 3: Denoise Human Faces

Examples
in the
database



Denoising
results



Noisy
(std:20)

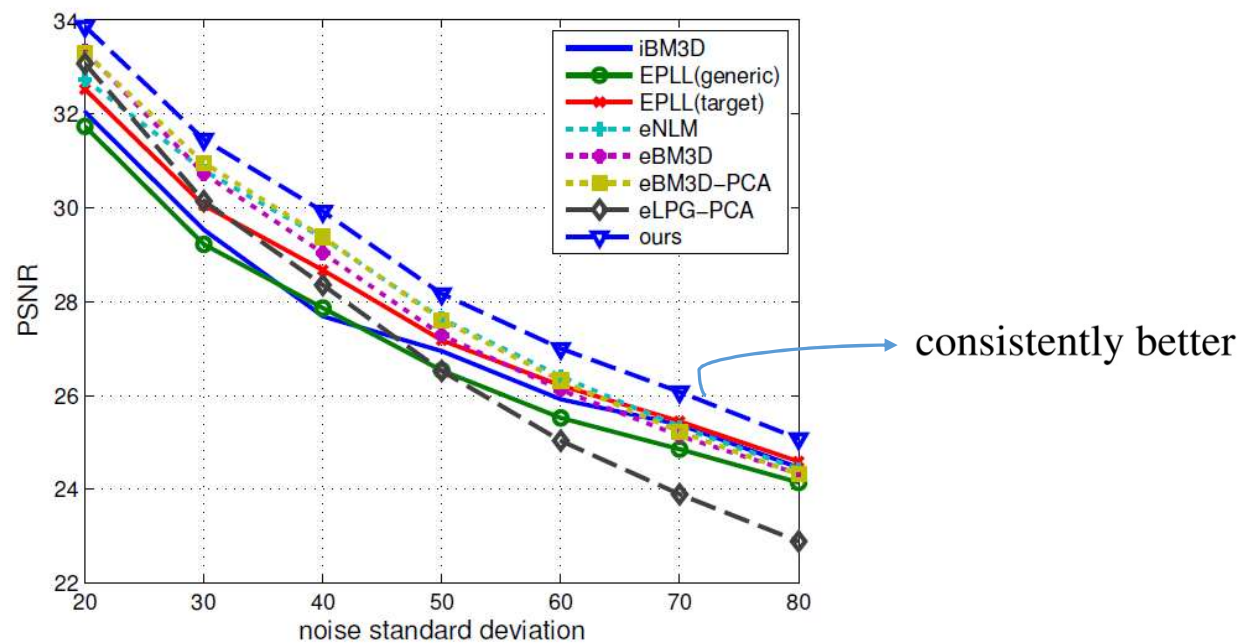
iBM3D
32.04dB

eNLM
32.74dB

eBM3D-PCA
33.29dB

ours
33.86dB

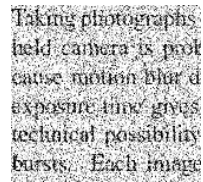
Experiment 3: Denoise Human Faces



Face denoising results: PSNR vs noise levels

Conclusion

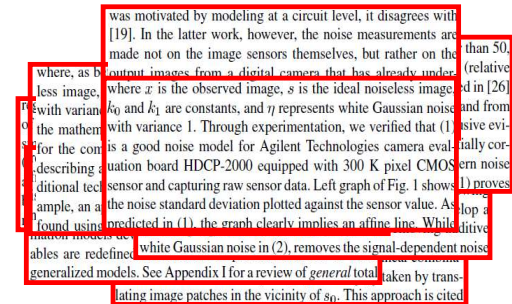
Conclusion 1: database adaptivity



noisy image



generic database

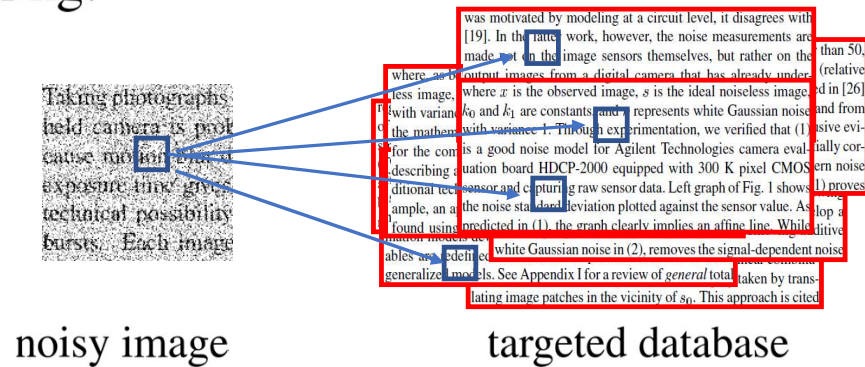


targeted database

Conclusion

Conclusion 2: algorithmic adaptivity

Patch matching:



Filter design:

Optimal denoising filter:

$$\hat{p} = U\Lambda U^T q$$

PCA of reference patches
 $\mathbf{p}_1, \dots, \mathbf{p}_k$

Wiener shrinkage based on
localized Bayesian prior

Future Work: Image Restoration



physical limitations
environmental factors



a degraded image
(with blur and noise, etc)



a sharp and clean image

image restoration

Thank you
Question?

Improving Λ

ℓ_α penalized BMSE:

$$\text{BMSE}_\alpha = \mathbb{E}_{\mathbf{p}} \left[\mathbb{E}_{\mathbf{q}|\mathbf{p}} \left[\left\| \mathbf{U} \Lambda \mathbf{U}^T \mathbf{q} - \mathbf{p} \right\|_2^2 \middle| \mathbf{p} \right] \right] + \gamma \|\Lambda \mathbf{1}\|_\alpha$$

Lemma 4 [Luo, Chan, and Nguyen 2014]: *Let s_i be the i th diagonal entry in $\mathbf{S}$, where $\mathbf{S}$ is the eigenvalue matrix of $\mathbf{P} \mathbf{W} \mathbf{P}^T$, then the optimal Λ that minimizes BMSE_α is $\text{diag} \{\lambda_1, \dots, \lambda_d\}$, where*

$$\lambda_i = \max \left(\frac{s_i - \gamma/2}{s_i + \sigma^2}, 0 \right), \quad \text{for } \alpha = 1,$$

and

$$\lambda_i = \frac{s_i}{s_i + \sigma^2} \mathbb{1} \left(\frac{s_i^2}{s_i + \sigma^2} > \gamma \right), \quad \text{for } \alpha = 0.$$

Overall Efficient Algorithm

Algorithm 1 Proposed Algorithm

Input: noisy patch $\mathbf{q}$, and similar patches $\mathbf{p}_1, \dots, \mathbf{p}_k$

Output: estimate $\hat{\mathbf{p}}$

Learn $\mathbf{U}$ and $\mathbf{\Lambda}$

- Form data matrix $\mathbf{P}$ and weight matrix $\mathbf{W}$
- Compute eigen-decomposition $[\mathbf{U}, \mathbf{S}] = \text{eig}(\mathbf{P}\mathbf{W}\mathbf{P}^T)$
- Compute $\lambda_i = \frac{s_i}{s_i + \sigma^2}$ or $\max\left(\frac{s_i - \gamma/2}{s_i + \sigma^2}, 0\right)$ or $\frac{s_i}{s_i + \sigma^2} \mathbb{1}\left(\frac{s_i^2}{s_i + \sigma^2} > \gamma\right)$

Denoise: $\hat{\mathbf{p}} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T \mathbf{q}$
